

- 4) A tip jar of loose coins contains twice as many quarters as dimes, three times as many nickels as pennies, and two fewer pennies than dimes. Altogether, the tip jar contains five dollars worth of coins. How many nickels are in the jar? (Hint: plug in the answers)

A) 12
B) 15
C) 18
D) 21

- 5) Three friends decide to split a pizza, dividing it up by the ratio 5 : 4 : 3, respectively. When the pizza arrives, it comes pre-cut into 8 equal slices. The first two friends get into a fight over how to distribute the pizza, but the third friend, after some quick math, simply takes their fair share. How many slices did the third friend take?

- 6) If $a : b = 2 : 3$ and $a : c = 1 : 4$, what is the value of $c : b$?

- 7) Mark has twice as many socks as shirts. Mark has five times as many scarves as socks. Mark has three times as many mittens as scarves. Mark has 120 mittens. Besides the mittens, how many other articles of clothing does Mark have?
- 8) A circular hula-hoop is sliced into three pieces. The first piece is painted red, while the second is painted blue. The third is painted green. The pieces are then re-assembled to form the original hula-hoop. The ratio of the different colored pieces is $r : b : g = 1 : 3 : 4$. If the radius of the hula hoop is 1 foot, what is the arc length of the green section of the hula hoop? (leave your answer in terms of π)
- 9) An aquarium likes to keep the ratio of larger fish (greater than 2 feet) to smaller fish (less than 1 foot) in a tank as close as possible to 1 : 5. One day, the aquarium stocks a new tank with 102 small fish and 15 large fish. The next day, the aquarium adds twice as many large fish as small fish to the same tank. How many large fish must have been added to make the proportion of large fish to small 1 : 5 ?

Answers:

- 1) 24 inches (or 2 feet)
- 2) $x = (2/3)z$
- 3) $3/2$
- 4) B (15)
- 5) 2 slices
- 6) 8 to 3
- 7) 52
- 8) π feet
- 9) 6 large fish